5.6 Shape 基类实现

```
class Shape {
public:
    ShapeType type;
    D2D1_COLOR_F color;
    bool selected;
    std::vector<D2D1_POINT_2F> points;
    bool filled;
    D2D1_COLOR_F fillColor;
    float lineWidth;
    LineStyle lineStyle;

    Shape(ShapeType t) : type(t), selected(false),
                         color(D2D1::ColorF(D2D1::ColorF::Black)),
                         filled(false), lineWidth(1.0f),
                         lineStyle(LineStyle::SOLID) {}
    virtual ~Shape() {}

    virtual void Draw(ID2D1RenderTarget* pRT,
                      ID2D1SolidColorBrush* pBrush) = 0;
    virtual bool HitTest(D2D1_POINT_2F point) = 0;
    virtual void Move(float dx, float dy) = 0;
    virtual D2D1_POINT_2F GetCenter() = 0;

    virtual void Fill(D2D1_COLOR_F color) {
        filled = true;
        fillColor = color;
    }

    virtual bool CanFill() const {
        return type == ShapeType::CIRCLE ||
               type == ShapeType::RECTANGLE ||
               type == ShapeType::TRIANGLE ||
               type == ShapeType::POLYLINE;
    }
};
```

Listing 12: Shape 基类定义